

Assignment No : 02

Roll no : 13216

Contents for Theory:1. Creation of Dataset using Microsoft Excel.3. Identification and Handling of Outliers

```
import pandas as pd

df=pd.read_csv("C:/Users/Kiran
Janjal/OneDrive/Desktop/student_performance.csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 49 entries, 0 to 48
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
0	math_score	44 non-null	float64
1	reading_score	45 non-null	object
2	writing_score	44 non-null	float64
3	performance_score	44 non-null	float64
4	club_joindate	47 non-null	float64

```
dtypes: float64(4), object(1)
```

```
memory usage: 2.0+ KB
```

```
df.fillna(23)
```

	math_score	reading_score	writing_score	performance_score
club_joindate				
0	69.0	81	63.0	100.0
2021.0				
1	68.0	81	72.0	98.0
2021.0				
2	62.0	84	60.0	77.0
2020.0				
3	63.0	76	63.0	75.0
2018.0				
4	72.0	75	72.0	23.0
2019.0				
5	74.0	78	67.0	23.0
2019.0				
6	56.0	80	76.0	23.0
2021.0				
7	67.0	80	60.0	23.0
2019.0				
8	3.0	79	60.0	23.0
2019.0				
9	23.0	77	79.0	86.0

2021.0				
10	23.0	79+B12:B23	78.0	91.0
23.0				
11	23.0	85	65.0	81.0
23.0				
12	23.0	84	74.0	85.0
2018.0				
13	23.0	83	23.0	93.0
2020.0				
14	60.0	77	23.0	96.0
2018.0				
15	78.0	82	23.0	95.0
551.0				
16	74.0	81	23.0	82.0
2021.0				
17	76.0	75	23.0	86.0
2018.0				
18	66.0	82	74.0	96.0
2018.0				
19	79.0	82	72.0	83.0
2019.0				
20	62.0	84	80.0	99.0
2020.0				
21	74.0	23	75.0	78.0
2020.0				
22	76.0	23	73.0	77.0
2019.0				
23	70.0	23	71.0	86.0
2020.0				
24	80.0	23	65.0	92.0
2021.0				
25	63.0	83	69.0	100.0
2021.0				
26	65.0	78	70.0	85.0
2021.0				
27	65.0	85	61.0	100.0
2020.0				
28	63.0	83	68.0	100.0
2020.0				
29	61.0	82	60.0	75.0
2021.0				
30	75.0	75	63.0	96.0
2020.0				
31	62.0	78	74.0	77.0
2019.0				
32	68.0	85	61.0	76.0
2020.0				
33	80.0	84	70.0	88.0
2018.0				

34	61.0	76	73.0	80.0
2021.0				
35	72.0	75	63.0	95.0
2021.0				
36	76.0	83	73.0	87.0
2018.0				
37	80.0	75	61.0	91.0
2021.0				
38	67.0	83	75.0	88.0
2020.0				
39	69.0	75	61.0	100.0
2020.0				
40	69.0	81	60.0	89.0
2019.0				
41	69.0	77	60.0	76.0
2020.0				
42	63.0	84	76.0	89.0
2018.0				
43	65.0	76	64.0	92.0
2019.0				
44	70.0	84	64.0	99.0
2018.0				
45	62.0	82	64.0	88.0
2019.0				
46	77.0	79	62.0	88.0
2021.0				
47	63.0	79	67.0	98.0
2020.0				
48	71.0	78	72.0	83.0
2018.0				

df.min

	writing_score	performance_score	math_score	reading_score
0	69.0	81	63.0	100.0
2021.0				
1	68.0	81	72.0	98.0
2021.0				
2	62.0	84	60.0	77.0
2020.0				
3	63.0	76	63.0	75.0
2018.0				
4	72.0	75	72.0	NaN
2019.0				
5	74.0	78	67.0	NaN
2019.0				
6	56.0	80	76.0	NaN
2021.0				
7	67.0	80	60.0	NaN

2019.0				
8	3.0	79	60.0	NaN
2019.0				
9	NaN	77	79.0	86.0
2021.0				
10	NaN	79+B12:B23	78.0	91.0
NaN				
11	NaN	85	65.0	81.0
NaN				
12	NaN	84	74.0	85.0
2018.0				
13	NaN	83	NaN	93.0
2020.0				
14	60.0	77	NaN	96.0
2018.0				
15	78.0	82	NaN	95.0
551.0				
16	74.0	81	NaN	82.0
2021.0				
17	76.0	75	NaN	86.0
2018.0				
18	66.0	82	74.0	96.0
2018.0				
19	79.0	82	72.0	83.0
2019.0				
20	62.0	84	80.0	99.0
2020.0				
21	74.0	NaN	75.0	78.0
2020.0				
22	76.0	NaN	73.0	77.0
2019.0				
23	70.0	NaN	71.0	86.0
2020.0				
24	80.0	NaN	65.0	92.0
2021.0				
25	63.0	83	69.0	100.0
2021.0				
26	65.0	78	70.0	85.0
2021.0				
27	65.0	85	61.0	100.0
2020.0				
28	63.0	83	68.0	100.0
2020.0				
29	61.0	82	60.0	75.0
2021.0				
30	75.0	75	63.0	96.0
2020.0				
31	62.0	78	74.0	77.0
2019.0				

32	68.0	85	61.0	76.0
2020.0				
33	80.0	84	70.0	88.0
2018.0				
34	61.0	76	73.0	80.0
2021.0				
35	72.0	75	63.0	95.0
2021.0				
36	76.0	83	73.0	87.0
2018.0				
37	80.0	75	61.0	91.0
2021.0				
38	67.0	83	75.0	88.0
2020.0				
39	69.0	75	61.0	100.0
2020.0				
40	69.0	81	60.0	89.0
2019.0				
41	69.0	77	60.0	76.0
2020.0				
42	63.0	84	76.0	89.0
2018.0				
43	65.0	76	64.0	92.0
2019.0				
44	70.0	84	64.0	99.0
2018.0				
45	62.0	82	64.0	88.0
2019.0				
46	77.0	79	62.0	88.0
2021.0				
47	63.0	79	67.0	98.0
2020.0				
48	71.0	78	72.0	83.0
2018.0>				

```
df.tail(5)
```

	math_score	reading_score	writing_score	performance_score
club_joindate				
44	70.0	84	64.0	99.0
2018.0				
45	62.0	82	64.0	88.0
2019.0				
46	77.0	79	62.0	88.0
2021.0				
47	63.0	79	67.0	98.0
2020.0				
48	71.0	78	72.0	83.0
2018.0				

```
df.head(5)
```

	math_score	reading_score	writing_score	performance_score
club_joindate				
0	69.0	81	63.0	100.0
2021.0				
1	68.0	81	72.0	98.0
2021.0				
2	62.0	84	60.0	77.0
2020.0				
3	63.0	76	63.0	75.0
2018.0				
4	72.0	75	72.0	NaN
2019.0				

```
df.describe()
```

	math_score	writing_score	performance_score	club_joindate
count	44.000000	44.000000	44.000000	47.000000
mean	67.386364	67.954545	88.545455	1988.382979
std	11.765570	6.175550	8.119613	214.224644
min	3.000000	60.000000	75.000000	551.000000
25%	63.000000	62.750000	82.750000	2019.000000
50%	68.500000	67.500000	88.000000	2020.000000
75%	74.000000	73.000000	96.000000	2021.000000
max	80.000000	80.000000	100.000000	2021.000000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 49 entries, 0 to 48
```

```
Data columns (total 5 columns):
```

#	Column	Non-Null Count	Dtype
---	-----	-----	-----
0	math_score	44 non-null	float64
1	reading_score	45 non-null	object
2	writing_score	44 non-null	float64
3	performance_score	44 non-null	float64
4	club_joindate	47 non-null	float64

```
dtypes: float64(4), object(1)
```

```
memory usage: 2.0+ KB
```

```
df
```

	math_score	reading_score	writing_score	performance_score
club_joindate				
0	69.0	81	63.0	100.0
2021.0				
1	68.0	81	72.0	98.0
2021.0				
2	62.0	84	60.0	77.0

2020.0				
3	63.0	76	63.0	75.0
2018.0				
4	72.0	75	72.0	NaN
2019.0				
5	74.0	78	67.0	NaN
2019.0				
6	56.0	80	76.0	NaN
2021.0				
7	67.0	80	60.0	NaN
2019.0				
8	3.0	79	60.0	NaN
2019.0				
9	NaN	77	79.0	86.0
2021.0				
10	NaN	79+B12:B23	78.0	91.0
NaN				
11	NaN	85	65.0	81.0
NaN				
12	NaN	84	74.0	85.0
2018.0				
13	NaN	83	NaN	93.0
2020.0				
14	60.0	77	NaN	96.0
2018.0				
15	78.0	82	NaN	95.0
551.0				
16	74.0	81	NaN	82.0
2021.0				
17	76.0	75	NaN	86.0
2018.0				
18	66.0	82	74.0	96.0
2018.0				
19	79.0	82	72.0	83.0
2019.0				
20	62.0	84	80.0	99.0
2020.0				
21	74.0	NaN	75.0	78.0
2020.0				
22	76.0	NaN	73.0	77.0
2019.0				
23	70.0	NaN	71.0	86.0
2020.0				
24	80.0	NaN	65.0	92.0
2021.0				
25	63.0	83	69.0	100.0
2021.0				
26	65.0	78	70.0	85.0
2021.0				

27	65.0	85	61.0	100.0
2020.0				
28	63.0	83	68.0	100.0
2020.0				
29	61.0	82	60.0	75.0
2021.0				
30	75.0	75	63.0	96.0
2020.0				
31	62.0	78	74.0	77.0
2019.0				
32	68.0	85	61.0	76.0
2020.0				
33	80.0	84	70.0	88.0
2018.0				
34	61.0	76	73.0	80.0
2021.0				
35	72.0	75	63.0	95.0
2021.0				
36	76.0	83	73.0	87.0
2018.0				
37	80.0	75	61.0	91.0
2021.0				
38	67.0	83	75.0	88.0
2020.0				
39	69.0	75	61.0	100.0
2020.0				
40	69.0	81	60.0	89.0
2019.0				
41	69.0	77	60.0	76.0
2020.0				
42	63.0	84	76.0	89.0
2018.0				
43	65.0	76	64.0	92.0
2019.0				
44	70.0	84	64.0	99.0
2018.0				
45	62.0	82	64.0	88.0
2019.0				
46	77.0	79	62.0	88.0
2021.0				
47	63.0	79	67.0	98.0
2020.0				
48	71.0	78	72.0	83.0
2018.0				

```
df.math_score.fillna(67)
```

0	69.0
1	68.0
2	62.0


```
3      63.0
4      72.0
5      74.0
6      56.0
7      67.0
8       3.0
9      67.0
10     67.0
11     67.0
12     67.0
13     67.0
14     60.0
15     78.0
16     74.0
17     76.0
18     66.0
19     79.0
20     62.0
21     74.0
22     76.0
23     70.0
24     80.0
25     63.0
26     65.0
27     65.0
28     63.0
29     61.0
30     75.0
31     62.0
32     68.0
33     80.0
34     61.0
35     72.0
36     76.0
37     80.0
38     67.0
39     69.0
40     69.0
41     69.0
42     63.0
43     65.0
44     70.0
45     62.0
46     77.0
47     63.0
48     71.0
Name: math_score, dtype: float64

df.columns.values
```

```
array(['math_score', 'reading_score', 'writing_score',  
      'performance_score', 'club_joindate'], dtype=object)
```

```
df.iloc[20]
```

```
math_score      62.0  
reading_score    84  
writing_score    80.0  
performance_score 99.0  
club_joindate   2020.0  
Name: 20, dtype: object
```

```
df.iloc[:,2:5]
```

	writing_score	performance_score	club_joindate
0	63.0	100.0	2021.0
1	72.0	98.0	2021.0
2	60.0	77.0	2020.0
3	63.0	75.0	2018.0
4	72.0	NaN	2019.0
5	67.0	NaN	2019.0
6	76.0	NaN	2021.0
7	60.0	NaN	2019.0
8	60.0	NaN	2019.0
9	79.0	86.0	2021.0
10	78.0	91.0	NaN
11	65.0	81.0	NaN
12	74.0	85.0	2018.0
13	NaN	93.0	2020.0
14	NaN	96.0	2018.0
15	NaN	95.0	551.0
16	NaN	82.0	2021.0
17	NaN	86.0	2018.0
18	74.0	96.0	2018.0
19	72.0	83.0	2019.0
20	80.0	99.0	2020.0
21	75.0	78.0	2020.0
22	73.0	77.0	2019.0
23	71.0	86.0	2020.0
24	65.0	92.0	2021.0
25	69.0	100.0	2021.0
26	70.0	85.0	2021.0
27	61.0	100.0	2020.0
28	68.0	100.0	2020.0
29	60.0	75.0	2021.0
30	63.0	96.0	2020.0
31	74.0	77.0	2019.0
32	61.0	76.0	2020.0
33	70.0	88.0	2018.0
34	73.0	80.0	2021.0

35	63.0	95.0	2021.0
36	73.0	87.0	2018.0
37	61.0	91.0	2021.0
38	75.0	88.0	2020.0
39	61.0	100.0	2020.0
40	60.0	89.0	2019.0
41	60.0	76.0	2020.0
42	76.0	89.0	2018.0
43	64.0	92.0	2019.0
44	64.0	99.0	2018.0
45	64.0	88.0	2019.0
46	62.0	88.0	2021.0
47	67.0	98.0	2020.0
48	72.0	83.0	2018.0

```
df.math_score.fillna(67, inplace=True)
```

C:\Users\Kiran Janjal\AppData\Local\Temp\

ipykernel_14544\1148490175.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace method.

The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.

```
df.math_score.fillna(67, inplace=True)
```

```
df['math_score'] = df['math_score'].fillna(67)
```

```
df.math_score.isna().sum()
```

```
np.int64(0)
```

```
df
```

	math_score	reading_score	writing_score	performance_score
club_joindate				
0	69.0	81	63.0	100.0
2021.0				
1	68.0	81	72.0	98.0
2021.0				
2	62.0	84	60.0	77.0
2020.0				
3	63.0	76	63.0	75.0
2018.0				

4	72.0	75	72.0	NaN
2019.0				
5	74.0	78	67.0	NaN
2019.0				
6	56.0	80	76.0	NaN
2021.0				
7	67.0	80	60.0	NaN
2019.0				
8	3.0	79	60.0	NaN
2019.0				
9	67.0	77	79.0	86.0
2021.0				
10	67.0	79+B12:B23	78.0	91.0
NaN				
11	67.0	85	65.0	81.0
NaN				
12	67.0	84	74.0	85.0
2018.0				
13	67.0	83	NaN	93.0
2020.0				
14	60.0	77	NaN	96.0
2018.0				
15	78.0	82	NaN	95.0
551.0				
16	74.0	81	NaN	82.0
2021.0				
17	76.0	75	NaN	86.0
2018.0				
18	66.0	82	74.0	96.0
2018.0				
19	79.0	82	72.0	83.0
2019.0				
20	62.0	84	80.0	99.0
2020.0				
21	74.0	NaN	75.0	78.0
2020.0				
22	76.0	NaN	73.0	77.0
2019.0				
23	70.0	NaN	71.0	86.0
2020.0				
24	80.0	NaN	65.0	92.0
2021.0				
25	63.0	83	69.0	100.0
2021.0				
26	65.0	78	70.0	85.0
2021.0				
27	65.0	85	61.0	100.0
2020.0				
28	63.0	83	68.0	100.0

2020.0				
29	61.0	82	60.0	75.0
2021.0				
30	75.0	75	63.0	96.0
2020.0				
31	62.0	78	74.0	77.0
2019.0				
32	68.0	85	61.0	76.0
2020.0				
33	80.0	84	70.0	88.0
2018.0				
34	61.0	76	73.0	80.0
2021.0				
35	72.0	75	63.0	95.0
2021.0				
36	76.0	83	73.0	87.0
2018.0				
37	80.0	75	61.0	91.0
2021.0				
38	67.0	83	75.0	88.0
2020.0				
39	69.0	75	61.0	100.0
2020.0				
40	69.0	81	60.0	89.0
2019.0				
41	69.0	77	60.0	76.0
2020.0				
42	63.0	84	76.0	89.0
2018.0				
43	65.0	76	64.0	92.0
2019.0				
44	70.0	84	64.0	99.0
2018.0				
45	62.0	82	64.0	88.0
2019.0				
46	77.0	79	62.0	88.0
2021.0				
47	63.0	79	67.0	98.0
2020.0				
48	71.0	78	72.0	83.0
2018.0				